Exam Date & Time: 25-Feb-2026 (01:15 PM - 03:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computer Science and Applications
University Theory Examination February- 2026
1st Semester of Master of Science in Information Technology (M.Sc. (IT))

MODERN SOFTWARE ENGINEERING [CTUC508]

Marks: 50 **Duration: 135 mins.**

Section - I

Answer all the questions. Section Duration: 20 mins

1 Apply your understanding of the Waterfall and identify the maincharacteristic of the Waterfall model.

1)	Overlapping phases		2)	Iterative development		3)	Sequential phases without overlap		4)	Prototype creation before coding	

(1)

Course Outcome: "CO1" / Cognitive Level: "Application"

2 Analyze the Waterfall model to determine why it is not suitable forprojects with rapidly changing requirements.

1)	It allows changes at any time		2)	It uses overlapping phases		3)	No flexibility for changes in requirements once development starts		4)	It requires user feedback after each phase	

(1)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

3 Apply your understanding for identify which document includes bothfunctional and non-functional requirements?

1)	SRS		2)	SDS		3)	Gantt Chart		4)	Test Plan	

(1)

Course Outcome: "CO2" / Cognitive Level: "Application"

4 Evaluate Which characteristic ensures each requirement can betraced to its origin?

1)	Verifiability		2)	Traceability		3)	Modifiability		4)	Completeness	

(1)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

5 Evaluate aggregation differs from composition because:

1)	Both are same		2)	Aggregation allows independent existence		3)	Composition allows independent existence		4)	None of the above	

(1)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

6 Evaluate which UML diagram represents best user interactions in anonline bookstore.

1)	Use Case Diagram		2)	Sequence Diagram		3)	Class Diagram		4)	Activity Diagram.	

(1)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

7 Apply your understanding How is Scrum different from Waterfall? (1)

1)	Scrum is iterative with feedback		2)	Waterfall is iterative		3)	Scrum uses sequential flow		4)	None	

Course Outcome: "CO4" / Cognitive Level: "Application"

8 Create an effective Agile workflow for fast-paced web appdevelopment.

1)	Pair Programming + Continuous Integration + Daily Stand-ups		2)	Sequential Design + Final Testing		3)	Long documentation + Annual Releases		4)	Limited Feedback	

(1)

Course Outcome: "CO4" / Cognitive Level: "Create"

9 Apply the DevOps pipeline to identify which stage checks real-timesystem performance.

1)	Build		2)	Code		3)	Monitor		4)	Release	

(1)

Course Outcome: "CO5" / Cognitive Level: "Application"

10 Develop a mini RMMM plan for an online portal that lists top 3risks and mitigation actions.

1)	Identify, Monitor, Manage		2)	Ignore, Delay, Retry		3)	Plan, Cancel, Close		4)	None	

(1)

Course Outcome: "CO5" / Cognitive Level: "Create"

Section-II

Answer 4 out of 6 questions.

1 Apply the Waterfall model steps and explain the process of softwaredevelopment.

(5)

Course Outcome: "CO1" / Cognitive Level: "Application"

2 Develop the process of Prototype Model and explain in detail.

(5)

Course Outcome: "CO1" / Cognitive Level: "Create"

3 Analyze the Functional and Non-Functional Requirements and explainthe difference between them.

(5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

4 Evaluate the concept of Reverse Engineering and explain objectives,goals, and steps.

(5)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

5 Draw Use Case Diagram for an Online Grocery Delivery Applicationthat allows users to order groceries online and have them deliveredto their location. Users must register and log in to place orders,and the system stores their details for future convenience. Adminscan manage products, track users, and process payments throughvarious payment gateways. The user can browse food categories, addgroceries to their cart, place orders, and pay online. Users cangenerate an invoice/bill for the groceries purchased. Admins havethe ability to update product information, add new items, and vieworders. (5)

Course Outcome: "CO3" / Cognitive Level: "Create"

6			Analyze the Core Values of Agile and explain in detail.	(5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

Section-III

Answer 4 out of 6 questions.

1			Create a Use Case Diagram for a Project Management System on thegiven scenario. Scenario: Project Management System For a company, which is, working on various kinds of projects,create a project management system that will be used to keep track of all the projects, employees, andmanagers with an individual login. Here manager will assign the respective projects to the employees. Oncethe project has been assigned to employees the notification will be generated and it will benotified to employee sand employees will go to create the project accordingly. Once the project will getcompleted, it will be updated in the system. The manager will see that as part of the report. The manager cangive feedback on the work. The manager will generate various reports such as Project completed, feedbackdetails, Ongoing Projects, etc.	(5)

Course Outcome: "CO3" / Cognitive Level: "Create"

2			Evaluate the DevOps Lifecycle and explain its key phases.	(5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

3			Evaluate the Agile Software Development methodology for developingsoftware.	(5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

4			Analyze the 7 Cs of DevOps and explain in detail.	(5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

5			Analyze the RMMM Plan and explain its components with examples.	(5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

6			Apply the core principles of DevOps to explain software developmentin detail.	(5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
University Exam Dec-Feb 2026	CTUC508:MODERN SOFTWARE ENGINEERING

#	Course Outcome	Description	Marks
1	CO1	Understanding Software Engineering Process Models.	12
2	CO2	Able to analyze requirements and mange project.	12
3	CO3	Get familiar with the concepts of Unified Modeling Language.	12
4	CO4	Understanding of the Agile software development concepts.	12
5	CO5	Able to apply DevOps concepts and understand risk management.	22

Total	**70**

* NA - CO Not Selected

Event Name	Subject
University Exam Dec-Feb 2026	CTUC508:MODERN SOFTWARE ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	6	14(20%)
4	Analysis	5	21(30%)
5	Create	5	17(24.29%)
6	Evaluate	6	18(25.71%)
	Total	22	70 (100%)